

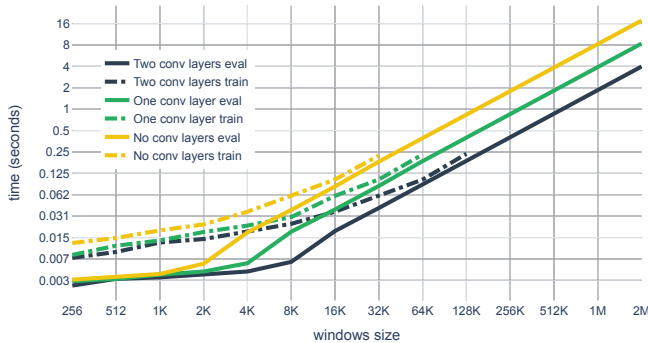


Figure 7: Training and evaluation speed (log-scale) depending on the prepended convolution count on the MusicNet task.

tions. We visualize the note predictions and describe the most common errors in Appendix ?? . Additional training details are provided in Appendix ?? .

5.4 Ablation study

We have chosen the multiplication task as a showcase for the ablation study. It is a hard task which challenges every aspect of the model and performance differences are clearly observable. We use a model with 2 Beneš blocks, 192 feature maps and train it on length 128. We consider the following simplifications of the proposed architecture:

- removing LayerNorm (without LayerNorm)
- using ReLU instead of GELU
- removing the residual connection; the last equation of RSU becomes $[o_1, o_2] = c$ (without residual)
- setting the residual weight parameter $\sigma(S)$ to a constant 1 instead of a learnable parameter; the equation becomes $[o_1, o_2] = i + h \odot c$ (without residual scale)

We can see in Fig. 8 that the proposed baseline performs the best. Versions without residual connection or normalization do not work well. Residual weight parameter and GELU non-linearity give a smaller contribution to the model’s performance.

We investigate the effect of the RSU hidden layer size on performance. Parameter count and speed of the model is directly proportional to the hidden layer size; therefore, we want to select the smallest size, which gives a good performance. By default, we use $2m$ feature maps where m is the number of feature maps of the model. Versions with half as large or twice as large hidden layer size are explored. We discover that a larger hidden layer size leads to better performance. We consider the choice of $2m$ hidden layer size a good compromise between performance and parameter count. The experimental results for various hidden layer sizes are found in Appendix ?? .

We have performed ablation experiments also for LAMBADA and MusicNet tasks with similar conclusions, but the differences are much less pronounced.

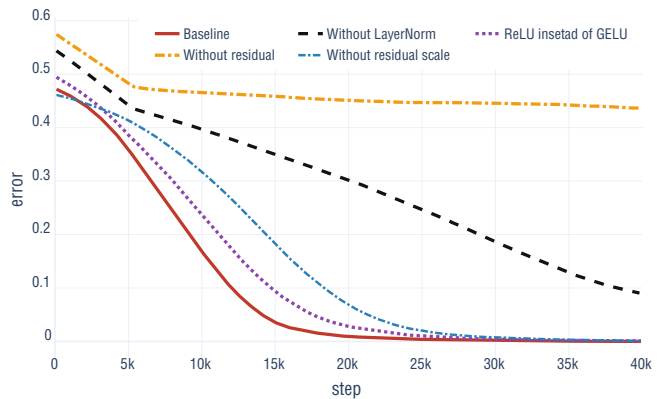


Figure 8: Ablation experiments. The plot shows test error on the multiplication task length 128 vs training step.

6 Conclusions

We have proposed a simpler and faster version of the neural Shuffle-Exchange network. It has $O(n \log n)$ complexity and enables processing of sequences up to length 2 million where standard methods, like attention, fail. While keeping the overall successful connectivity structure of the Shuffle-Exchange network, we have shown that using residual connections instead of gated connections in its design, gives a significant boost to the training speed and achieved accuracy. Additionally, we have shown how to combine the model with strided convolutions that increase its speed and sequence length that can be processed.

The proposed model achieves state-of-the-art accuracy in recognizing musical notes directly from the waveform – a task where the ability to process long sequences is crucial. Notably, our architecture uses significantly fewer parameters than most of the previous best models for this task.

Our experiments confirm the Residual Shuffle-Exchange network as a useful building block for long sequence processing applications.

Acknowledgements

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